

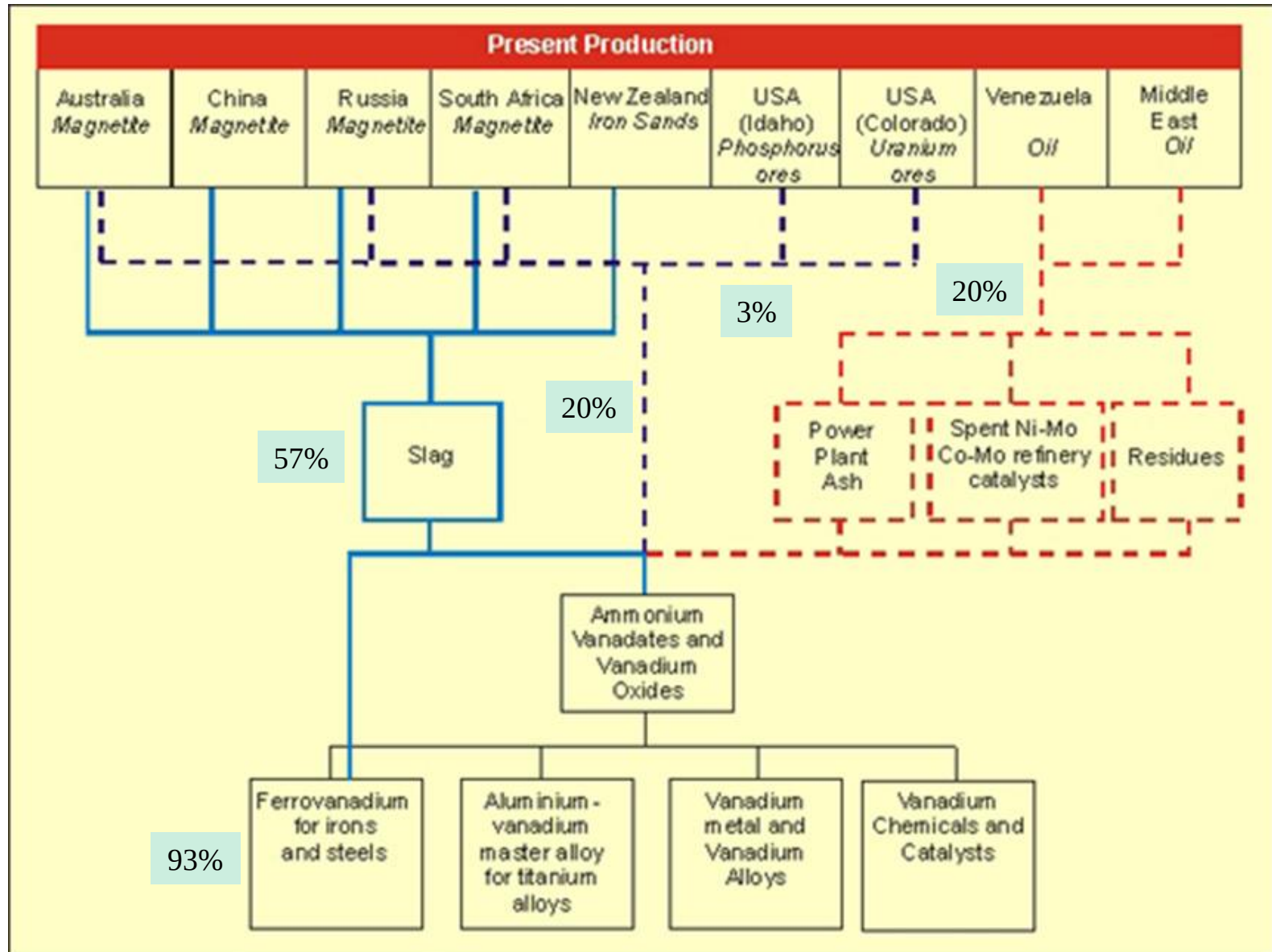
Recovery of Vanadium from LD-Slag – 50 years of Swedish Experiences

Guozhu Ye, Swerea MEFOS
guozhu.ye@swerea.se

Presentation

- Vanadium worldwide and Nordic perspective
- Vanadium research in Sweden, 5 decades
- V-balance, problem or opportunity
- VILD project, a possible solution?

Vanadium Sources and Production



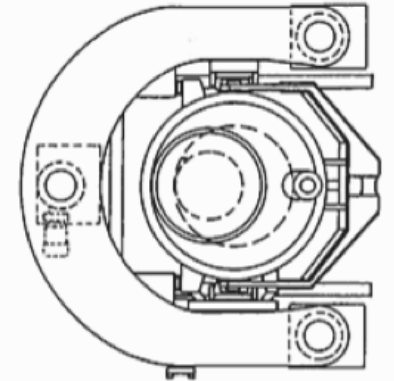
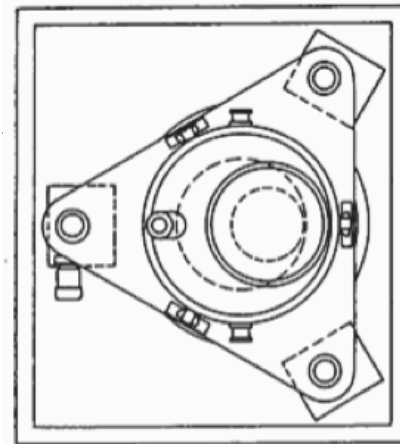
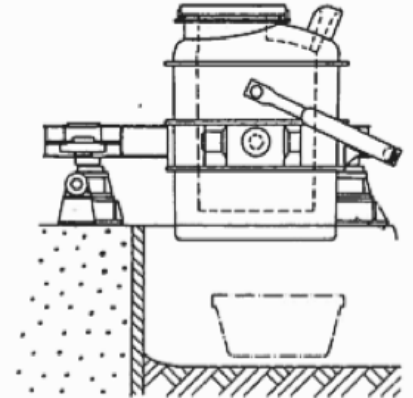
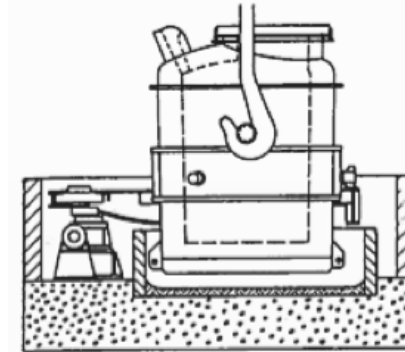
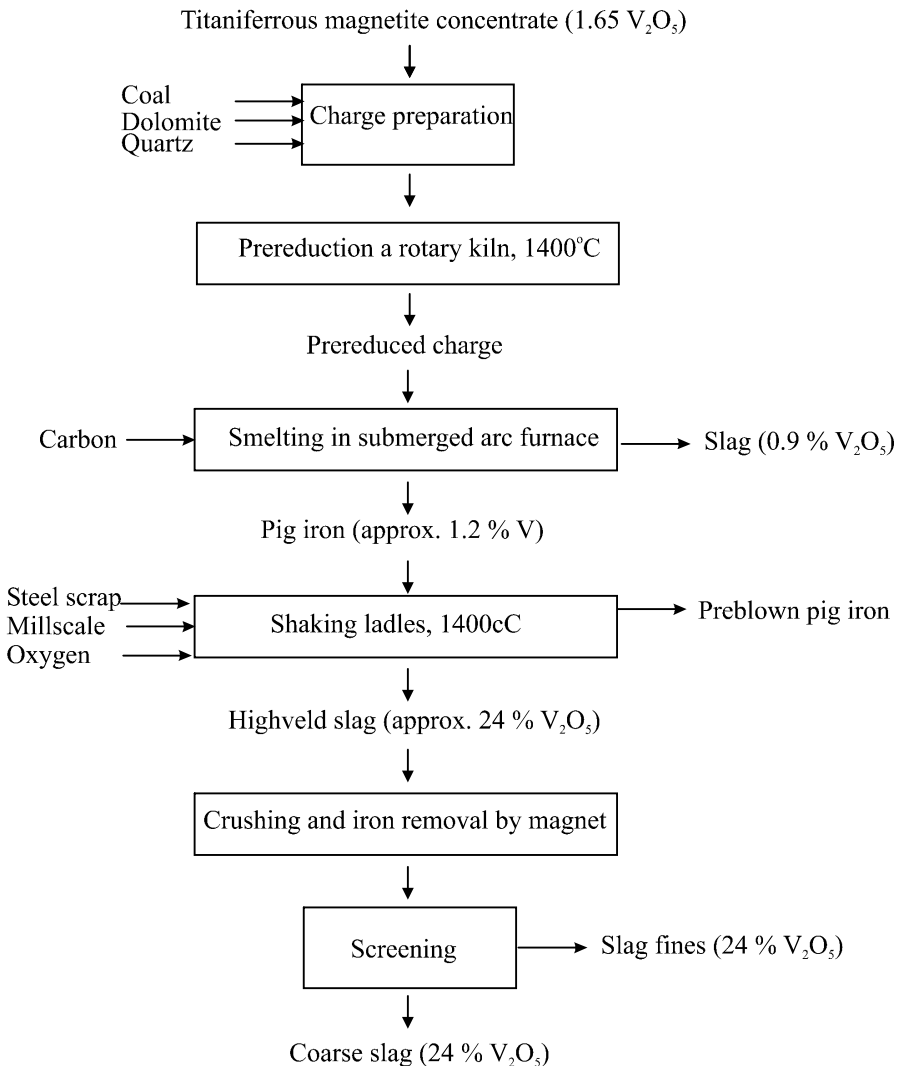
The major V-slag producers

- Highveld, SA – Shaking ladle
- Pan-Steel, China – Double converter
- EVRAZ NTMK, Russia – Duplex Process (Nizhniy Tagit)
- New Zealand Steel – Transfer ladle

Process principle

- Preoxidation for V-removal
- Principle: $\underline{V} + O_2 = V_2O_3$
- Mill scale and oxygen for oxidation, scrap for cooling

The Highveld process, South Africa



a

b

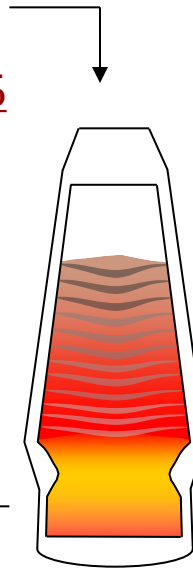
Fig. 4. Skakskänk med Δ -ram, a), och med hästskoram, b). Enl. Tunder och Hölle.²²

The V-process at Pan-steel, China

- Production of 13 000 ton V-slag (17%V₂O₅)

Pan-iron ore:

-13%TiO₂,
-0.55-0.7%V₂O₅
- 43-54%Fe



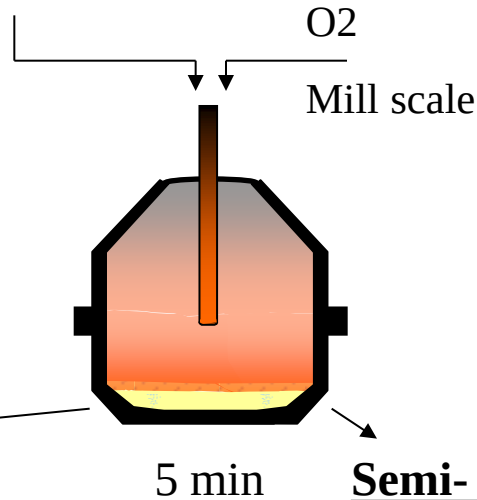
BF-slag: 22%TiO₂

Slag/Hm = 0.67

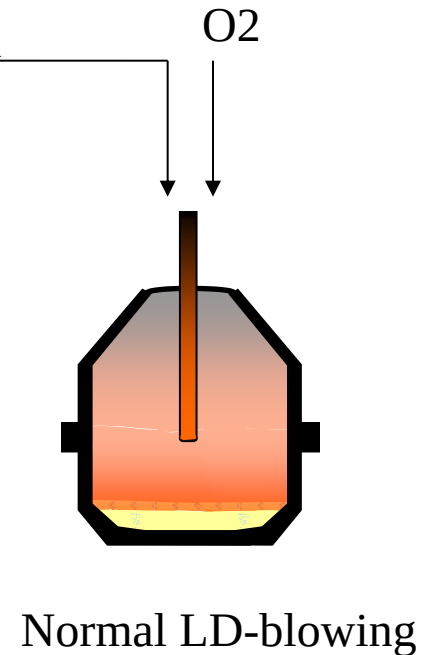
V-slag:

-17%V₂O₅,
-14%SiO₂,
-9%Fe-metall,
-26%FeO,
-5-7%MnO,
-0.01%P₂O₅

Hot metal: 0.3%V, 0.08%P, 0.2-0.3%Si

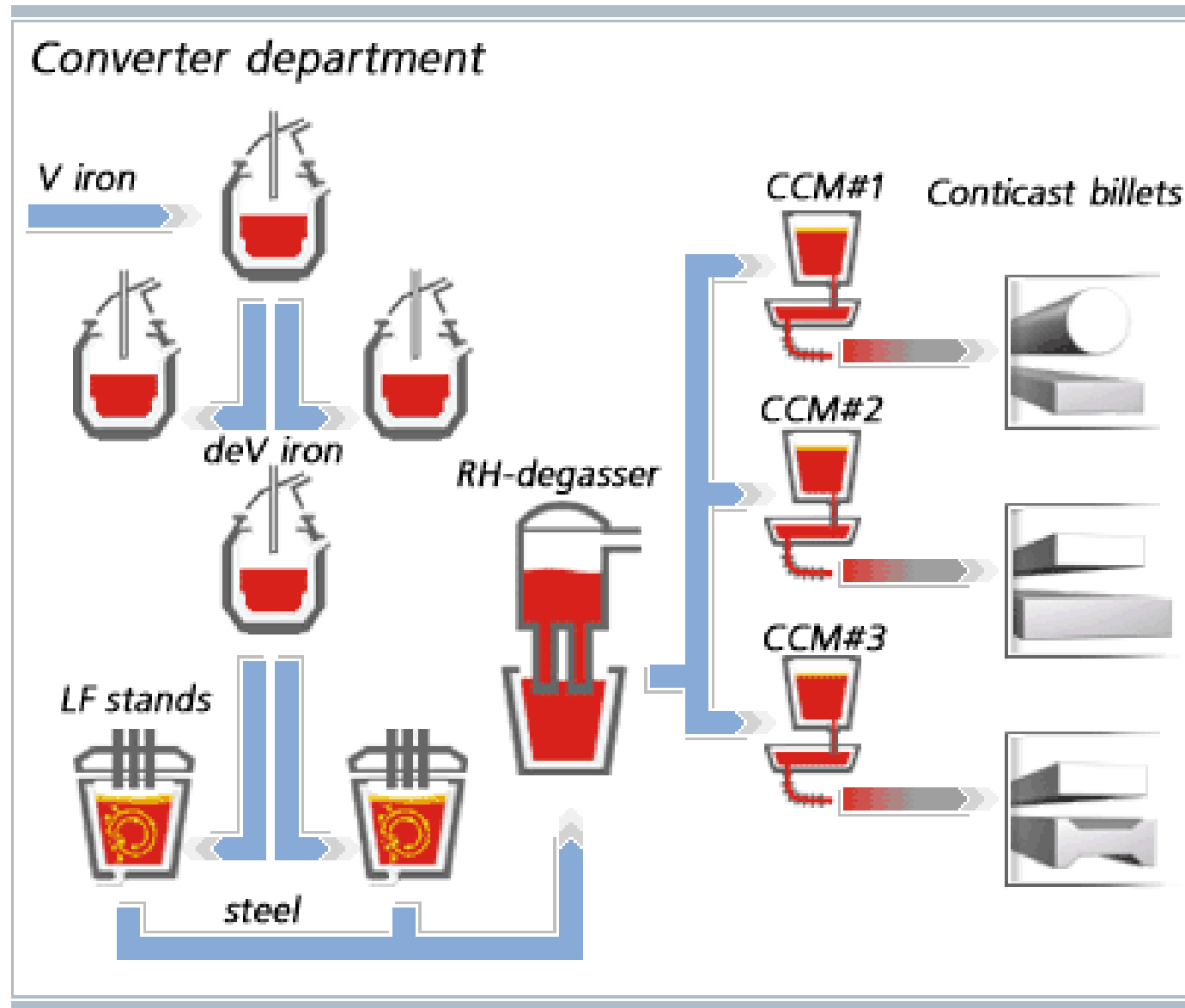


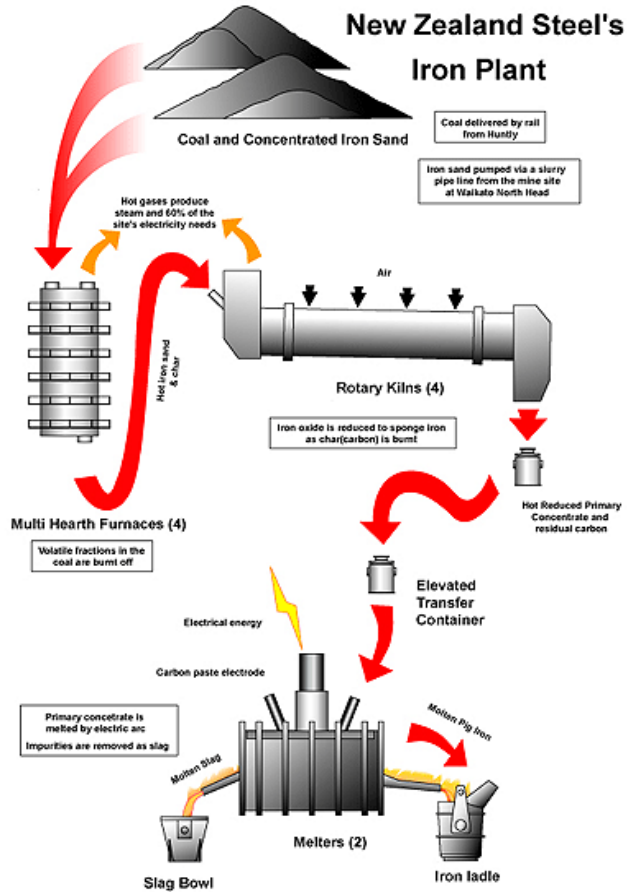
Semi- steel
with 0.06%V



Normal LD-blowing

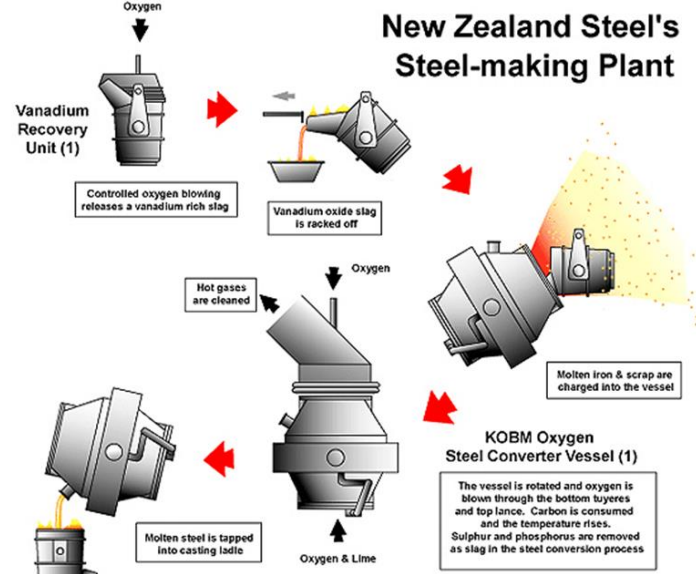
The NTMK-Duplex process, Russia





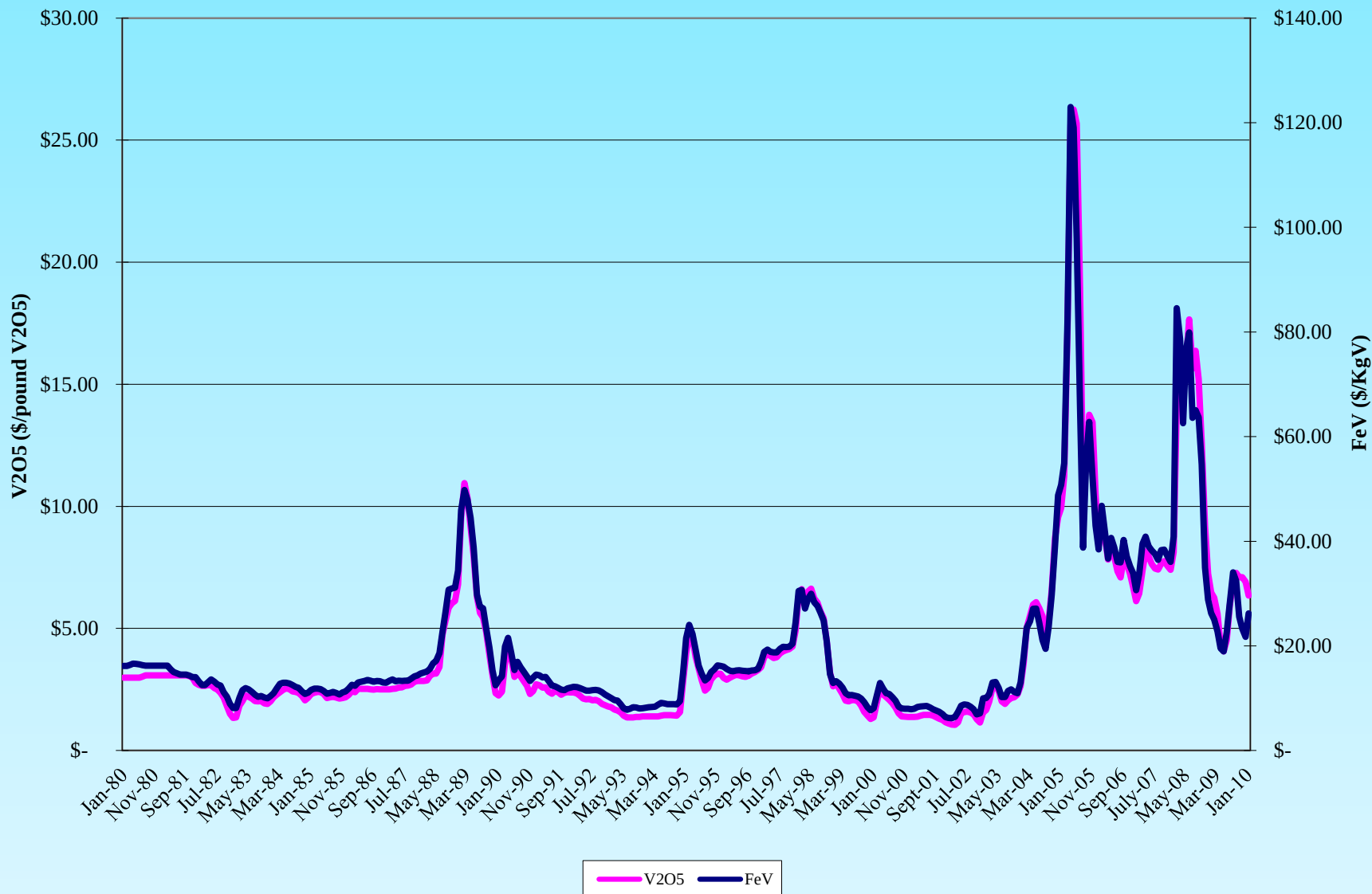
Analysis of pig iron (%):

Fe	95.50
C	3.40
Si	0.11
Mn	0.33
P	0.06
S	0.06
<u>V</u>	<u>0.42</u>
Ti	0.16

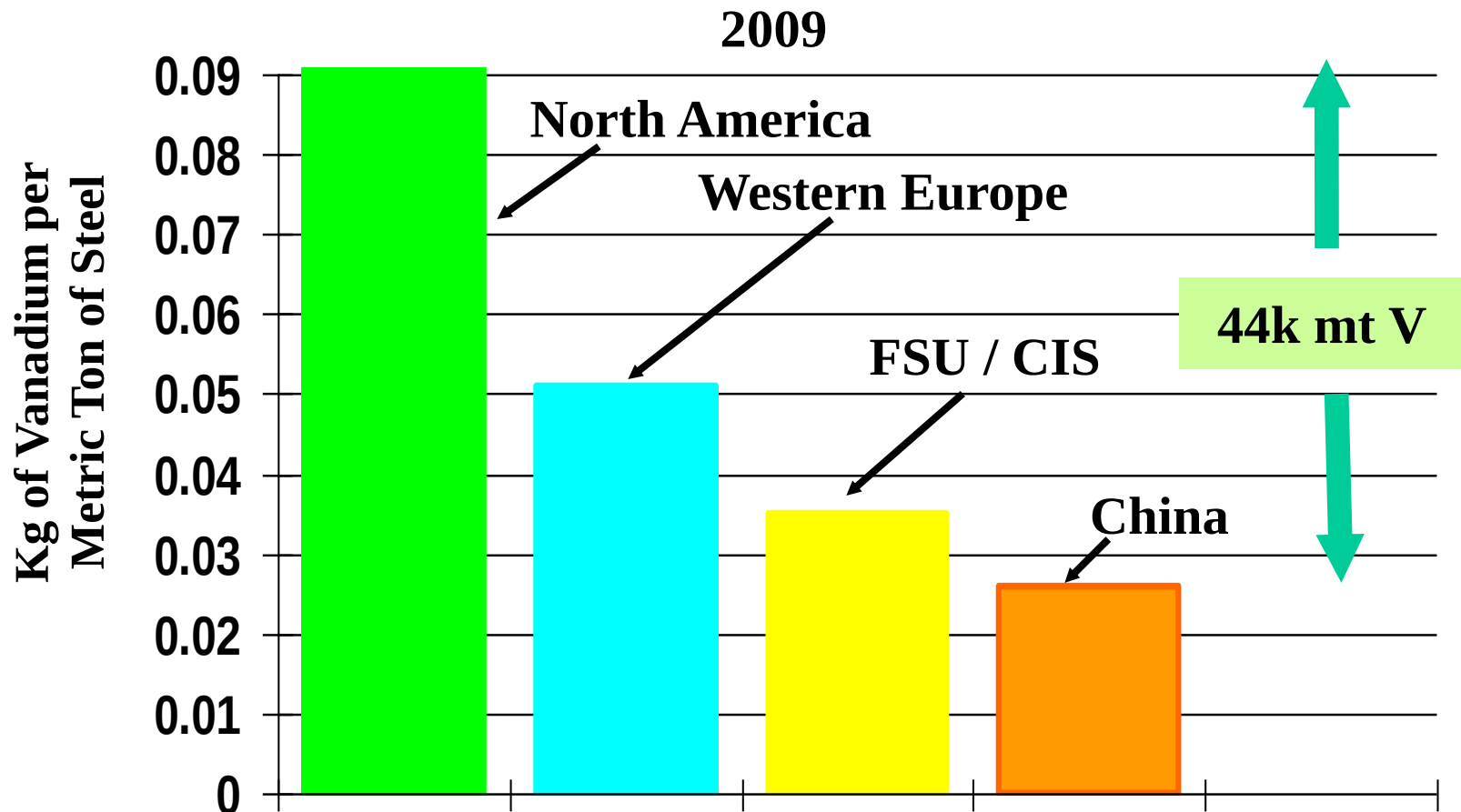


30 Years of V2O5 and FeV Price History

(London Metal Bulletin monthly midpoint averages)



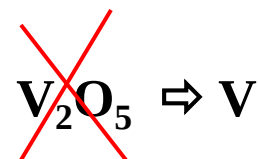
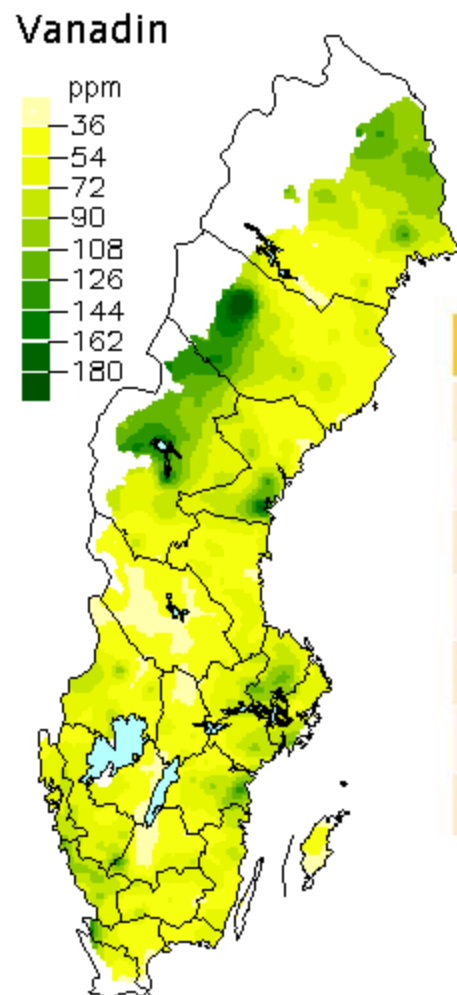
Vanadium Usage per Ton of Steel



Vanadium processes, Nordic countries

- ELKEM Bremanger process
- Rautaruukki Mustavaarra
Vanadium

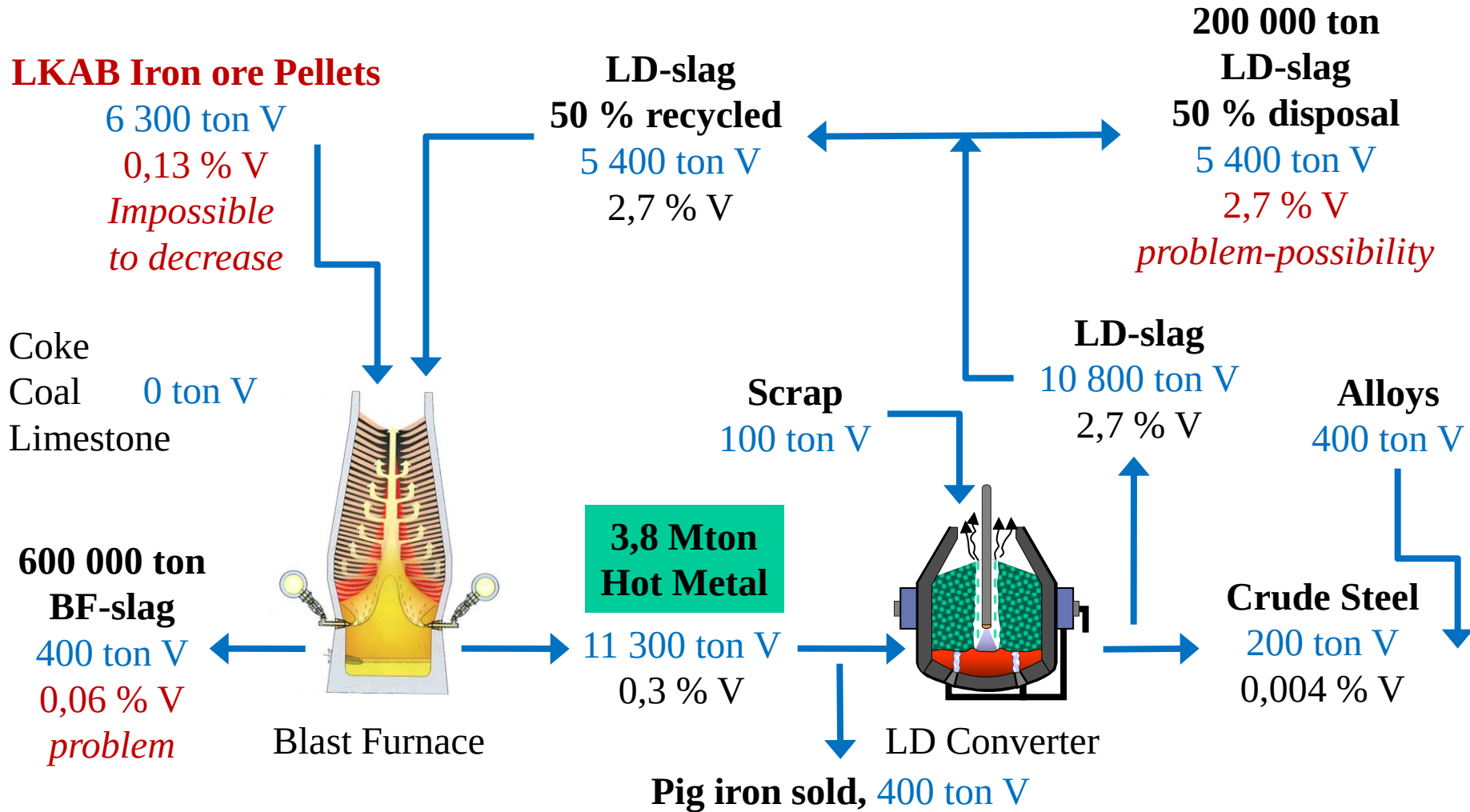
Vanadium in the Swedish soils



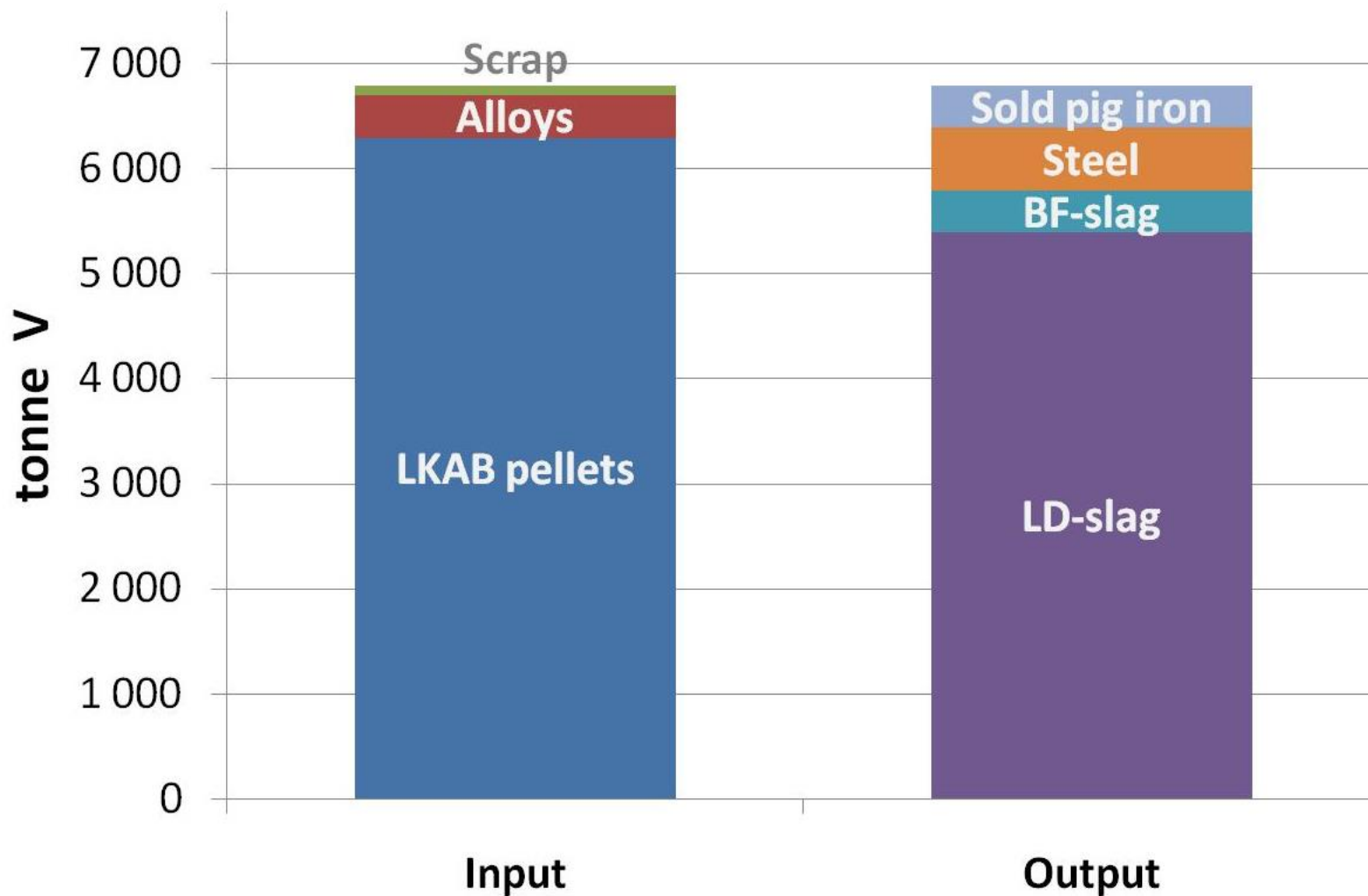
	ppm V
Limit for contaminated soil used in sensitive ground – today	120
Limit for contaminated soil used in sensitive ground – "tomorrow"	100
Limit for contaminated soil used in <u>less</u> sensitive ground – "tomorrow"	200
Vanadium in the earth crust – average value	120
Vanadium in ordinary cement	50
Vanadium in blast furnace slag	600
Vanadium in LD-slag (BOF-slag)	27 000

<http://www-markinfo.slu.se/sve/kem/totkem/v.html>

Vanadium balance at SSAB 2007



Vanadium balance, SSAB 2007



Purposes

High V
LD-slag



FeV-product

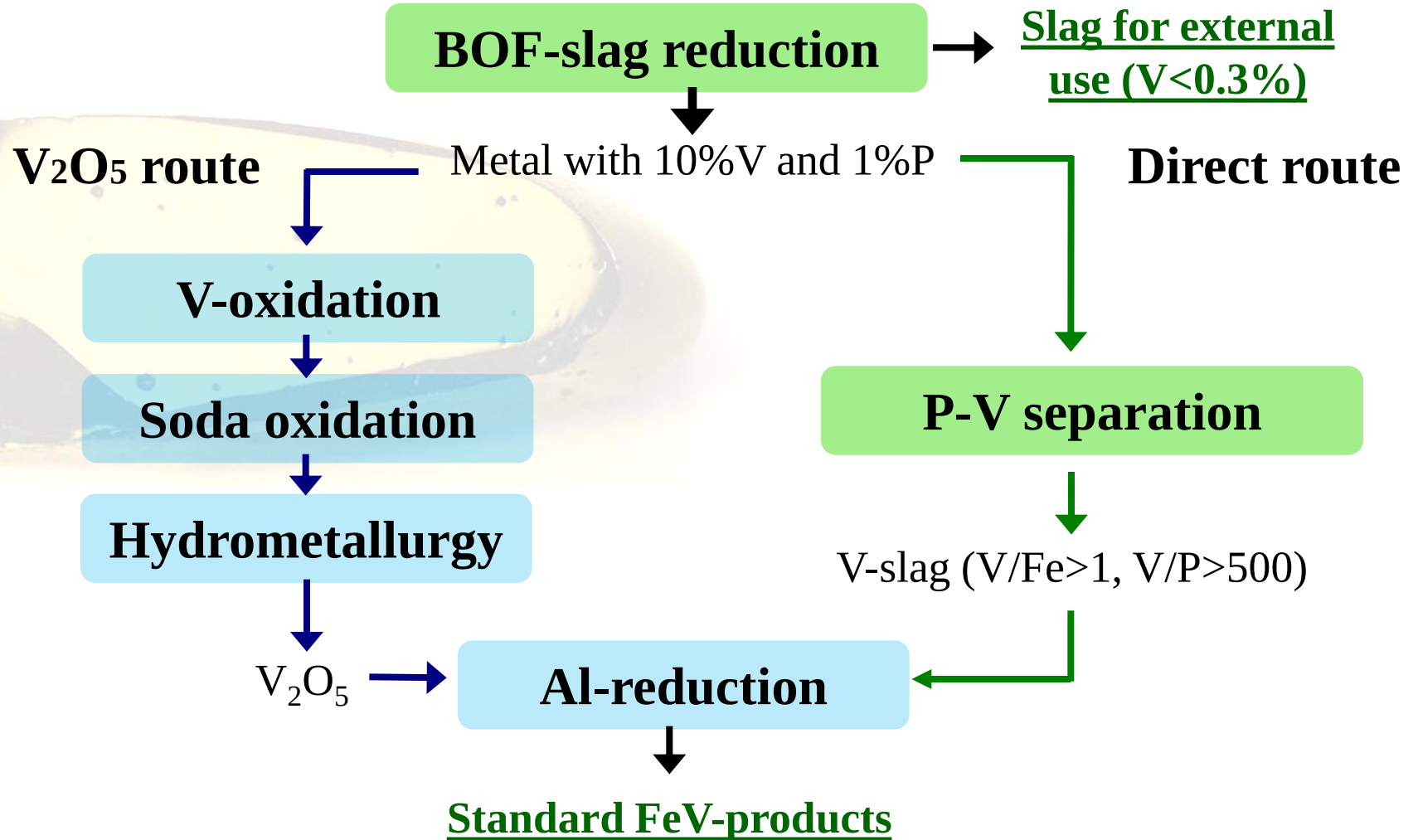


Slag product

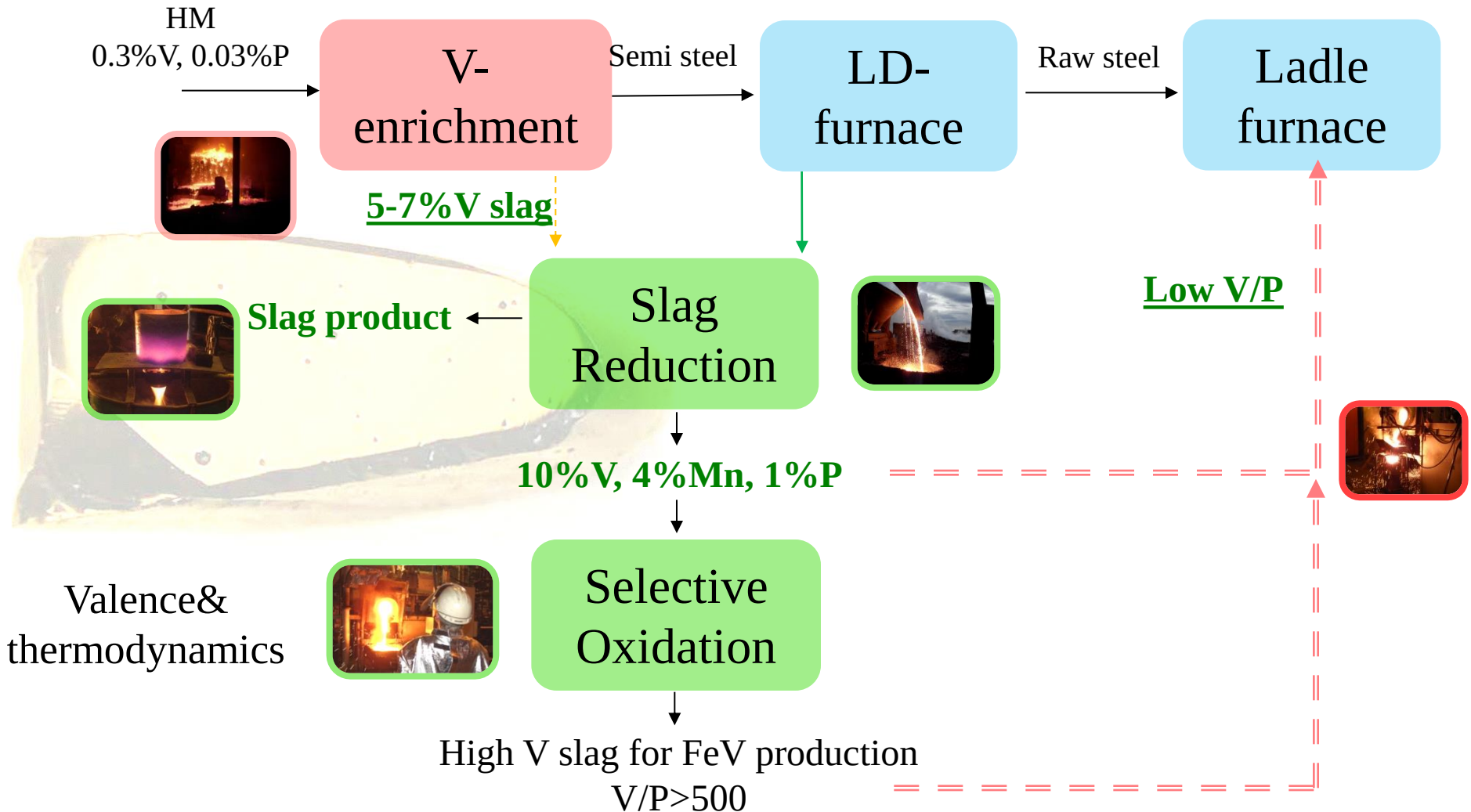


- A cost efficient and environmentally sound processing concept
- Two products, zero wastes
- World leading V-research group

VILD Phase I, 2005-2008



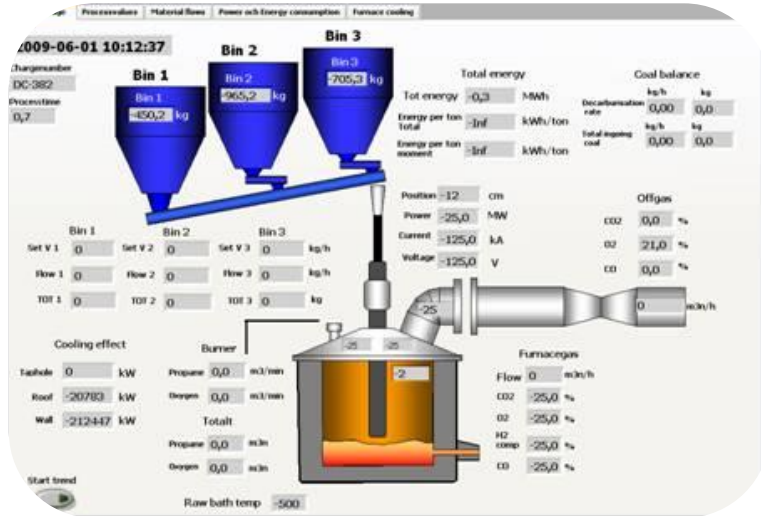
VILD Phase II, 2009-2012



PRISMA: Consequences

Project by the industrial partners: V in soil, SLU, HTH

Reduction of cold and hot LD-slag



Reduction in EAF: 2 campaigns in 2008-2009



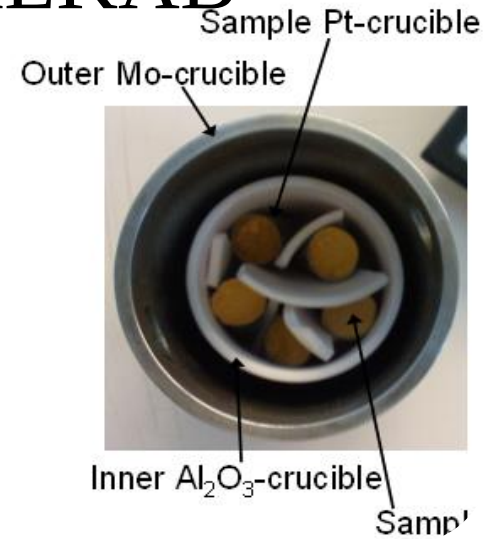
mass%	Fe	V	CaO	SiO ₂	MnO	P ₂ O ₅	Al ₂ O ₃	MgO
Slag 1	0.08	0.03	41.63	37.40	0.32	0.01	2.90	17.20
Slag 2	0.07	0.03	37.62	14.17	0.21	0.01	37.92	11.22
Slag 3	0.64	0.04	37.20	23.53	0.34	0.01	27.86	11.92

Slag Product

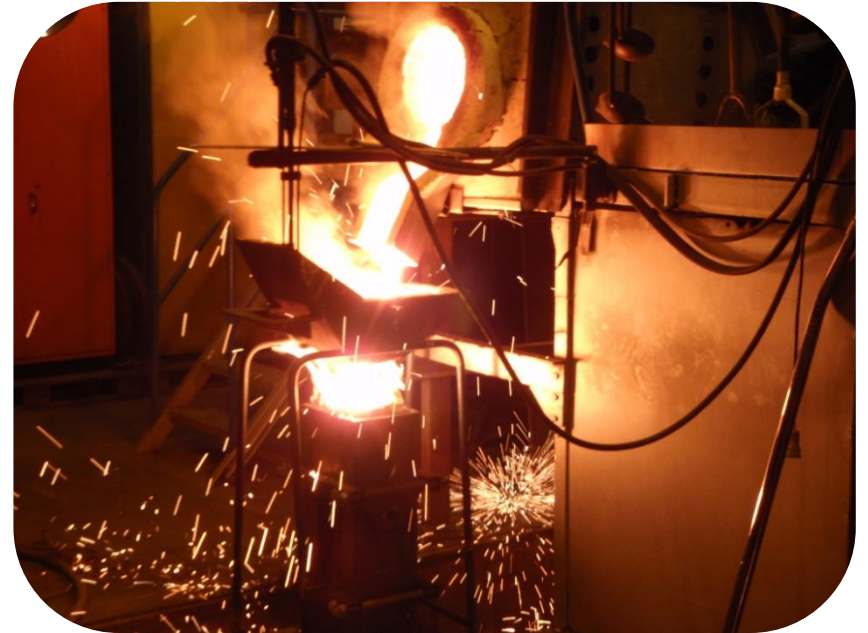
- Slag from AC- and DC-trials have been granulated, crushed and mixed with OPC
- Strength measured after 2, 28 and 91 days
- MERIT5000 + OPC as reference

	R2162	D413	MERIT5000/OPC
2d	12.8	15.6	16.5
28d	43.6	54.4	52.2
91d	56.7	60.2	61.0
Glashalt	73.2	97.3	99
Densitet	3018	2961	2924
Blaine	460	466	505

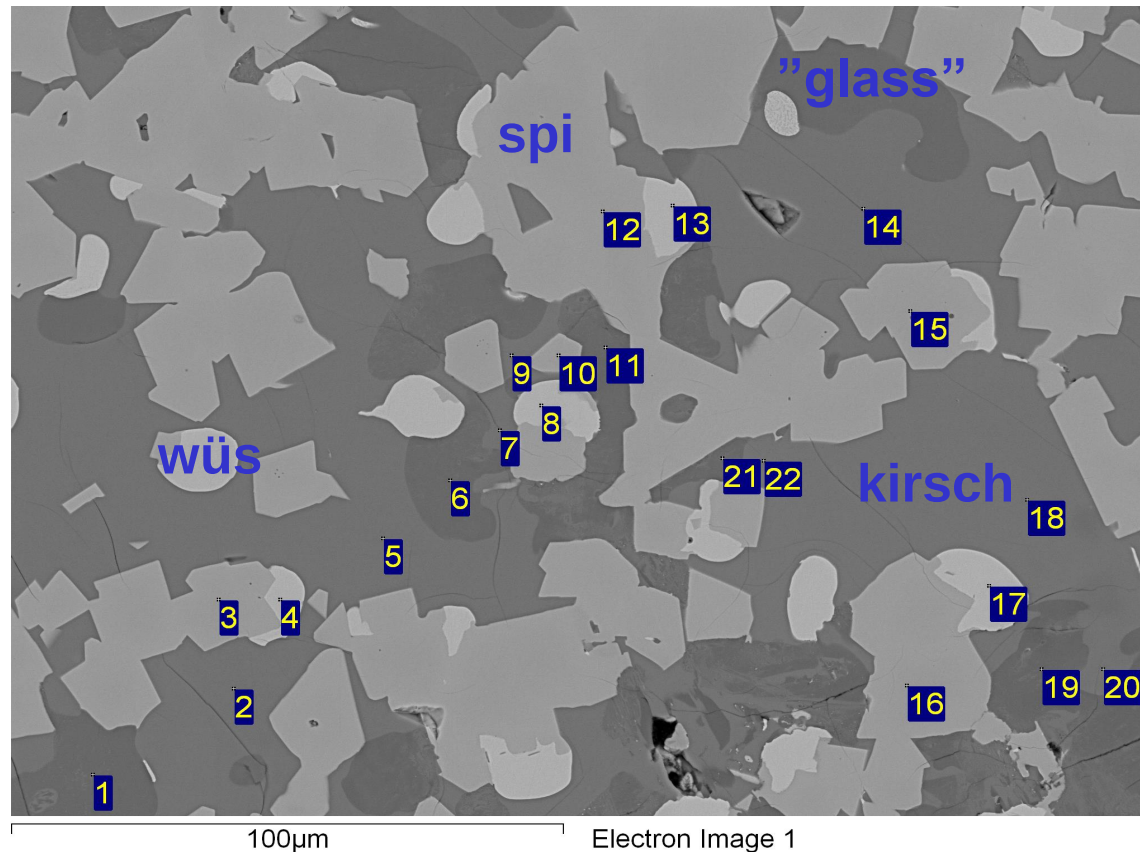
Selective Oxidation, MEFOS&LKAB



Can a FeV product with higher P-content be used directly? KIMAB



Studied at Ruukki



	MgO	Al ₂ O ₃	SiO ₂	P ₂ O ₅	CaO	TiO ₂	V ₂ O ₅	Cr ₂ O ₃	MnO	FeO	Total
spinel	0.7	0.6	0.2	0.0	0.5	4.5	34.9	15.3	10.9	38.0	105.6
kirschsteinite	1.3	0.1	33.8	0.4	34.2	0.6	0.5	0.2	11.6	18.4	101.1
"glass"	0.3	0.1	31.2	5.4	52.1	0.2	1.3	0.3	4.2	5.8	100.8
wüstite	0.8	0.1	0.3	0.0	0.5	0.3	1.6	0.3	15.6	80.5	100.1

Summary - Genreal

Mineral Processing (70-80's)

- Cooling control
- Spinell with high vanadium ($\text{FeO} \cdot \text{V}_2\text{O}_3$)
- High V in the C2S phase ($\text{CaO-SiO}_2\text{-V}_2\text{O}_5\text{-P}_2\text{O}_5$)

Preoxidation (80's)

- Comprehensive project
- V/Si the key issue

Direct processes (90's-)

- Direct acid leaching (89%)
- Direct soda roasting – leaching (High soda consumption)

Slag reduction(2000-)

- Cold or hot slag
- Zero waste

Summary - VILD

Reduction

- Cold or hot slag
- High V-recovery
- Slag could be used in cement
- What reactor best?

Selective oxidation

- 150 kg scale OK
- Low V solubility/slag viscosity
- Valence and thermodynamics

Several products

- Basic high V-slag
- FeV (10-20%V, 1%P)
- Acid high V-slag
- Direct standard FeV

Important factors

- Right customers who need V-bearing raw materials and will pay for the V-products!
- Flexible products – minimum risks!!
- Timing and the V-market
- Close colaboration with V-producers, long term partnership, visions for development
- Environmental drivers!!!!